

YUJIAN LIU

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RESEARCH

LLM post-training, reinforcement learning for LLM agents, agentic systems and inference, LLM alignment

EDUCATION

University of California, Santa Barbara, Santa Barbara, CA Sep 2022 – Dec 2026 (expected)
Ph.D. student in Computer Science, advised by **Prof. Shiyu Chang**

University of Michigan, Ann Arbor, Ann Arbor, MI Sep 2018 – May 2022
M.S. in Computer Science, advised by **Prof. Lu Wang** and **Prof. Rada Mihalcea**
B.S. in Computer Science, Minor in Mathematics
GPA: 4.00/4.00

Shanghai Jiao Tong University, Shanghai, China Sep 2016 – Aug 2018
B.S. in Electrical and Computer Engineering (Dual Degree)
GPA: 3.75/4.00

WORK EXPERIENCE

LinkedIn CoreAI, Mountain View, CA Feb 2026 – Sep 2026
Research Intern

- Built an **agentic auto-research framework** that autonomously explores LLM architecture variants, discovering **linear attention architectures** whose pretrained models outperform GDN2.
- Developed **online KV-cache compaction** for long-horizon LLM agents using **token eviction, attention matching, and self-generated proxy queries**, reducing KV cache by 80% while retaining most task performance and improving inference throughput.
- Developed **self-speculating agents** that reuse the prefix KV cache to predict and pre-execute future tool calls; trained with **joint agent-speculator RL** on on-policy rollouts, improving tool-call prediction while preserving task success.

AMD GenAI, Santa Clara, CA Jun 2025 – Sep 2025
Research Intern

- Developed a **VLM-based inference-time learning framework** for computer-use agents, distilling online videos into structured UI-action trajectories and dynamically retrieving them as **in-context demonstrations**; improved success rate by +2.1 points on OSWorld-Verified and +4.2 points on WebArena.

MIT-IBM Watson AI Lab, Cambridge, MA Jun 2024 – Sep 2024
Research Intern

- Developed Prereq-Tune, a two-stage **SFT** framework that disentangles knowledge acquisition from task-skill learning and uses **synthetic data** to align fine-tuning with model-internal knowledge, reducing **LLM hallucinations** across short- and long-form generation.

Adobe Research, San Jose, CA Jun 2023 – Sep 2023
Research Intern

- Developed a **knowledge-augmented table QA** framework that uses LLMs to retrieve and synthesize external knowledge into expanded tables, followed by **SQL-based program reasoning**; outperformed strong baselines on WikiTableQuestions, TAT-QA, and FinQA.

SELECTED PUBLICATIONS

* Equal contribution; † Equal advising

- [1] **Yujian Liu***, Jiabao Ji*, Li An*, Rohit Jain, Gungor Polatkan, Siyu Zhu, Shiyu Chang. “Practical Online KV Cache Compaction for LLM Agents: An Empirical Study”. **arXiv, 2026**. [paper].
- [2] Jiabao Ji*, **Yujian Liu***, Li An*, Rohit Jain, Gungor Polatkan, Siyu Zhu, Shiyu Chang. “Speculate While You Reason: Teaching Agents to Predict Their Next Tool Call via Joint Agent-Speculator RL”. **EMNLP-Findings, 2026**. [paper].
- [3] **Yujian Liu***, Jiabao Ji*, Li An*, Tommi Jaakkola[†], Yang Zhang[†], Shiyu Chang[†]. “How well do agentic skills work in the wild: Benchmarking llm skill usage in realistic settings”. **COLM, 2026**. [paper] [code].
- [4] **Yujian Liu**, Ze Wang, Hao Chen, Ximeng Sun, Xiaodong Yu, Jialian Wu, Jiang Liu, Emad Barsoum, Zicheng Liu, Shiyu Chang. “Learning from Online Videos at Inference Time for Computer-Use Agents”. **TMLR, 2026**. [paper] [code].
- [5] Bairu Hou, Yang Zhang, Jiabao Ji, **Yujian Liu**, Kaizhi Qian, Jacob Andreas, and Shiyu Chang. “ThinkPrune: Pruning Long Chain-of-Thought of LLMs via Reinforcement Learning”. **TMLR, 2026**. [paper] [code].
- [6] **Yujian Liu**, Jiabao Ji, Yang Zhang, Wenbo Guo, Tommi Jaakkola, Shiyu Chang. “HarnessLLM: Automatic Testing Harness Generation via Reinforcement Learning”. **DL4C@NeurIPS, 2025**. [paper] [code].
- [7] Ziyang Jiang*, Li An*, **Yujian Liu**, Jiabao Ji, Qiucheng Wu, Jacob Andreas[†], Yang Zhang[†], Shiyu Chang[†]. “VISUALSKILL: Multimodal Skills for Computer-Use Agents”. **arXiv, 2026**. [paper].
- [8] **Yujian Liu**, Shiyu Chang, Tommi Jaakkola, and Yang Zhang. “Fictitious Synthetic Data Can Improve LLM Factuality via Prerequisite Learning”. **ICLR, 2025**. [paper] [code].
- [9] **Yujian Liu**, Jiabao Ji, Tong Yu, Ryan Rossi, Sungchul Kim, Handong Zhao, Ritwik Sinha, Yang Zhang, and Shiyu Chang. “Augment before You Try: Knowledge-Enhanced Table Question Answering via Table Expansion”. **EMNLP-Findings, 2025**. [paper] [code].
- [10] **Yujian Liu**, Yang Zhang, Tommi Jaakkola, and Shiyu Chang. “Correcting Diffusion Generation through Resampling”. **CVPR, 2024, Highlight**. [paper] [code].
- [11] Qiucheng Wu*, **Yujian Liu***, Handong Zhao, Trung Bui, Zhe Lin, Yang Zhang, and Shiyu Chang. “Harnessing the Spatial-Temporal Attention of Diffusion Models for High-Fidelity Text-to-Image Synthesis”. **ICCV, 2023**. [paper] [code].
- [12] Qiucheng Wu, **Yujian Liu**, Handong Zhao, Ajinkya Kale, Trung Bui, Tong Yu, Zhe Lin, Yang Zhang, and Shiyu Chang. “Uncovering the Disentanglement Capability in Text-to-Image Diffusion Models”. **CVPR, 2023**. [paper] [code].
- [13] Bairu Hou, **Yujian Liu**, Kaizhi Qian, Jacob Andreas, Shiyu Chang, and Yang Zhang. “Decomposing Uncertainty for Large Language Models through Input Clarification Ensembling”. **ICML, 2024, Oral** [paper] [code].
- [14] **Yujian Liu**, Yang Zhang, Tommi Jaakkola, and Shiyu Chang. “Revisiting Who’s Harry Potter: Towards Targeted Unlearning from a Causal Intervention Perspective”. **EMNLP, 2024, Oral**. [paper] [code].
- [15] Jiabao Ji, **Yujian Liu**, Yang Zhang, Gaowen Liu, Ramana Rao Kompella, Sijia Liu, and Shiyu Chang. “Reversing the Forget-Retain Objectives: An Efficient LLM Unlearning Framework from Logit Difference”. **NeurIPS, 2024**. [paper] [code].
- [16] Li An, **Yujian Liu**, Yepeng Liu, Yang Zhang, Yuheng Bu, and Shiyu Chang. “Defending LLM Watermarking Against Spoofing Attacks with Contrastive Representation Learning”. **COLM, 2025**. [paper] [code].

Full publication list: Google Scholar

TEACHING AND MENTORSHIP

- **Teaching Assistant** at University of California, Santa Barbara Fall 2025, 2024, 2023, 2022
Introduction to Deep Learning, Machine Learning, Introduction to Optimization
- **Teaching Assistant** at University of Michigan, Ann Arbor Winter 2022, 2021
Introduction to Natural Language Processing
- **Mentor** in Early Research Scholars Program (University of California, Santa Barbara) 2022, 2023
Project: Uncertainty in large language models, automatic prompt generation for diffusion models

SELECTED AWARDS

UCSB Graduate Division, Graduate Dissertation Fellowship Award: June 2026
James B. Angell Scholar, University of Michigan: March 2020